Ubiquitous Intelligence for 6G Security: Hierarchical Federated Small Language Models for Resilient False Base Station Detection*

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Abstract

As the industry transitions toward 6G and Beyond 5G (B5G) systems, the paradigm is shifting from simple connectivity to ubiquitous intelligence. However, this evolution introduces significant security vulnerabilities, specifically the persistent threat of False Base Stations (FBS) that execute identity catching, Man-in-the-Middle (MitM) attacks, and service rejections. Traditional centralized AI solutions often fail to meet the strict privacy mandates and resource constraints of edge devices. This paper proposes a decentralized framework for on-device FBS detection that integrates Federated Learning (FL) with a Small Language Model (SLM) classifier built on a hierarchical packet-encoder bridge: a learned residual MLP maps each Radio Resource Control (RRC) and Non-Access Stratum (NAS) packet to a virtual token, which is concatenated with a short text instruction prefix and fed to a frozen 4-bit NormalFloat (NF4) quantized 270-million parameter Gemma-3 base with trainable Low-Rank Adaptation (LoRA) adapters. To enable federation under realistic edge constraints, we introduce Hierarchical Federated Adaptation (HFA), in which the packet encoder is personalized per client to absorb device-level feature variation while LoRA adapters are aggregated across clients via FedAvg, yielding a privacy-by-design layer that never transmits raw signaling logs. We evaluate the framework on the MODI cellular signaling corpus across a non-IID per-file partition of 110 clients. Preliminary pilot results across all four regimes show that federated LoRA-only training matches the centralized upper bound (test F1 0.850 vs. 0.834, accuracy 0.833 vs. 0.818) while reducing per-round communication by approximately one order of magnitude relative to federating the full bf16 base, and that HFA additionally shifts the operating point toward fewer false alarms (FPR 0.275 vs. 0.278–0.284 for the other regimes) at parity F1.

1 Introduction

As the industry moves toward 6G and Beyond 5G (B5G) systems, the research focus has shifted from technical capacity to the development of secure, highly intelligent networks. While the 5G era enabled significant growth in mobile communication across its core service categories, it was largely an evolutionary step in terms of overall network data rates. 6G intends to revolutionize this field by targeting massive connectivity, ultra-high speeds exceeding 100 Gbps, and sub-millisecond latency [1, 2, 3, 4]. Following the 2023 ITU-R M.2160 recommendation [5], the

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industry has embraced ubiquitous intelligence as a primary design requirement. These systems are expected to drive autonomous network behavior and context-aware logic, serving as the critical backbone for the digital services and smart cities of the future.

The ongoing shift toward Beyond 5G (B5G) and 6G systems is accompanied by heightened security vulnerabilities, most notably the persistent threat of False Base Stations (FBS). These rogue entities, often referred to as fake base stations, impersonate legitimate network nodes to manipulate User Equipment (UE) into performing unauthorized handovers or initial attachments. Once a connection is established, an FBS can orchestrate various malicious activities, including eavesdropping, IMSI catching, Man-in-the-Middle (MitM) interceptions, and denial-of-service attacks, all of which compromise user confidentiality and network integrity. Recent global incidents underscore the gravity of this threat. In early 2025, a mobile FBS in Bangkok transmitted fraudulent SMS phishing links to over 30,000 individuals in public spaces, resulting in several significant arrests [6, 7]. Similar operations involving SMS blaster devices have been dismantled in the UK, Qatar, Indonesia, and Turkey [8]. These instances demonstrate the expanding complexity and international reach of FBS operations. As B5G networks grow in density and mobility, the ability to detect and neutralize these threats requires sophisticated security frameworks and real-time intelligence.

To mitigate these risks, the integration of Artificial Intelligence (AI) and Machine Learning (ML) is becoming central to network defense [9, 10, 11, 12]. According to 3GPP Release 19 [13], AI/ML is now a fundamental pillar of the next-generation Radio Access Network (RAN), particularly for tasks such as channel state information (CSI) estimation, beam management, and high-accuracy UE positioning. These advancements facilitate localized, data-driven logic essential for responding to FBS threats instantaneously. However, standard ML models [9, 10, 11] are frequently constrained by their inability to manage the scale of B5G security requirements, often struggling with user privacy mandates and the heavy overhead of centralized processing.

The current dominance of Large Language Models (LLMs) further complicates the deployment of Federated Learning (FL) due to the massive computational requirements of models featuring billions of parameters. This scale introduces critical constraints on communication bandwidth and memory capacity. Under limited bandwidth conditions, transmitting complete model updates is often unfeasible, while the limited memory of edge devices makes it difficult to process these extensive models for synchronization. Addressing these hurdles requires a shift toward resource-efficient strategies that maintain the high-level intelligence necessary for 6G security.

The necessity for a decentralized intelligence architecture stems from the logistical and regulatory difficulties of modern data management. Traditional centralized AI paradigms require aggregating massive volumes of raw signaling data from millions of UEs at a single location. In the context of 6G, this approach is increasingly untenable due to the sheer volume of traffic and the resulting strain on uplink bandwidth. Furthermore, stringent data privacy regulations and the sensitive nature of network telemetry often prevent users and operators from sharing raw data. Because FBS attacks are typically localized and transient, centralized systems can also suffer from response delays. A collaborative learning framework is therefore required to extract intelligence from distributed edge data without the liabilities of data centralization.

This research introduces a paradigm shift by combining FL with Small Language Models (SLMs) to facilitate FBS detection at the network edge. The FL framework allows multiple UEs to collectively train a global model by exchanging only local model updates rather than sensitive raw logs, ensuring a privacy-by-design approach. We use SLMs, efficient transformer-based architectures, to reason over Radio Resource Control (RRC) and Non-Access Stratum (NAS) signaling as structured packet sequences. Unlike their larger counterparts, SLMs are specifically

designed for the limited computational and memory envelopes of 5G/6G edge devices. To overcome the token-budget ceiling that arises when serializing wide protocol-field sets as text, we replace per-packet text serialization with a learned residual-MLP *packet encoder* that projects each packet into the SLM's hidden space as a single virtual token, allowing the SLM to attend over a fixed, short context window regardless of the number of underlying fields.

The primary contributions of this paper are as follows:

- We design a hierarchical SLM detector that pairs a learned per-packet encoder with a frozen 4-bit quantized base model and trainable LoRA adapters, enabling on-device FBS classification at constant SLM context length irrespective of the underlying RRC/NAS field count.
- We introduce *Hierarchical Federated Adaptation* (HFA), a decomposition that personalizes the packet encoder per client to absorb device-level feature variation while federating only the LoRA adapters, reducing per-round communication while preserving cross-device protocol reasoning.
- We define a leakage-clean evaluation protocol with file-level splits and hash-deterministic placement of singleton-scenario captures, and evaluate four training regimes (centralized, federated full adapters, federated LoRA-only, and HFA) on the MODI cellular signaling corpus.

2 Background and Related Works

The detection of FBS is critical for ensuring the security of mobile networks, particularly in the context of emerging 5G and future 6G technologies. As a result, FBS detection has become a prominent area of research in mobile network security. Various methodologies have been proposed to address this issue, including network-based detection, specification-based approaches, machine learning techniques, and cryptographic solutions.

Nakarmi et al. [14] introduced a network-based detection system for 3GPP technologies that operate without requiring user-installed software, leveraging network topology and configuration information for more reliable detection. Their system was successfully implemented and validated in both laboratory settings and real-world trials, leading to its adoption by the 3GPP standardization organization. FBSleuth [15] presents a forensic framework that utilizes unique radio frequency fingerprints to accurately identify and associate FBS devices with criminal activities. Bolcek et al. [16] use In-phase and Quadrature (IQ) data of a radio frequency (RF) signal to identify a device working as FBS. This method enhances security by effectively detecting FBSs in flexible, multi-vendor network setups.

SMDFbs [10] is a behavior rule specification-based FBS detection system that derives behavior rules from the normal operations of base stations and converts these rules into a state machine. Based on this state machine, the system detects network anomalies and mitigates threats. Addressing the root vulnerability of FBSs, a hierarchical identity-based signature protocol called Schnorr-HIBS [17] was proposed to provide efficient broadcast authentication with minimal overhead. The protocol outperforms existing 3GPP authentication methods by achieving over six times faster cryptographic delays and reducing communication costs by 31%.

Machine learning techniques have been applied to detect FBSs [9, 18]. By utilizing robust features based on Reference Signals Received Power (RSRP) and various ML algorithms, such as Regression Clustering, Anomaly Detection Forest, Autoencoder, RCGAN, and XGBoost, several approaches have shown high precision in detection even when the false base station is

using a legitimate PCI. Furthermore, Mubasshir et al. [19] propose FBSDetector, an effective and efficient detection solution that can reliably identify FBSes. Unlike many network-side solutions, FBSDetector operates on the UE side, leveraging layer-3 network traces and applying ML models to uncover malicious behaviors. Closer to the present work, GAN-based augmentation has been used to expand limited FBS datasets [20], and quantum-enhanced classifiers have been explored as a complementary detection front-end [21]; both motivate the need for richer model architectures that can leverage the structural grammar of cellular signaling.

Cellular Control-Plane Vulnerabilities. The threat model for FBS detection is shaped by a sustained line of work exposing weaknesses in 4G/5G control-plane protocols. Practical attacks against LTE privacy and availability [22], systematic adversarial testing via model checking (LTEInspector [23]; 5GReasoner [24]) and dynamic security analysis [25] have catalogued unauthenticated RRC/NAS message types that FBSes exploit. Identity-mapping, payload modification, and DNS redirection on the data link layer [26], full UE impersonation in 4G [27], AKA-protocol privacy threats spanning 3G/4G/5G [28], side-channel attacks on the paging protocol [29], and vulnerabilities in the handover procedure [30] together establish that FBS exploitation is not a single attack but a spectrum of normative violations distributed across RRC and NAS message exchanges. Our framework targets the union of these violations by treating an entire short window of signaling – rather than a single message – as the unit of detection.

Device-Side IMSI-Catcher Detection. A parallel line of work focuses on user-equipment-side detection without requiring network-operator cooperation. Early IMSI-catcher-catcher prototypes [31] demonstrated feasibility, and tools such as SnoopSnitch [32] and AIMSICD [33] brought the approach to commodity Android devices. SeaGlass [34] extended detection to city-wide passive monitoring, while PHOENIX [35] formalized device-centric protocol monitoring through runtime verification. More recent systems including CellGuard [36] and CellDAM [37] push toward rootless, user-space deployments. Our approach inherits this UE-side deployment target but replaces hand-crafted rules and feature engineering with a learned, on-device SLM detector that does not require root or kernel modifications because it operates on already-collected baseband diagnostic logs.

Protocol-as-Language Baselines. The notion of treating structured network traffic as a token sequence consumable by a language model originates outside cellular security. DeepLog [38] and LogBERT [39] apply sequence models to system logs for anomaly detection; ET-BERT [40] adapts BERT-style pretraining to encrypted-traffic datagrams; and netFound [41] proposes a foundation model for general network security. These works validate the protocol-as-language premise but use either bespoke pretraining or raw-text serialization, which scales poorly to the hundreds-of-fields-per-packet regime characteristic of RRC/NAS captures. Our hierarchical packet-encoder bridge directly addresses this scaling barrier by replacing the field-level text serialization with a learned dense projection.

Federated Learning in Next-Generation RANs. Surveys of AI in future wireless systems consistently identify federated learning as a primary enabler for privacy-preserving, on-device intelligence in 5G and 6G [42, 43, 44]. The architectural assumption is that raw traffic and telemetry should not leave the UE; only model updates should. Existing FBS-detection literature has not explored this regime end-to-end on a realistic non-IID per-device partition. We bridge this gap by treating each captured signaling file as a federated client, exposing the

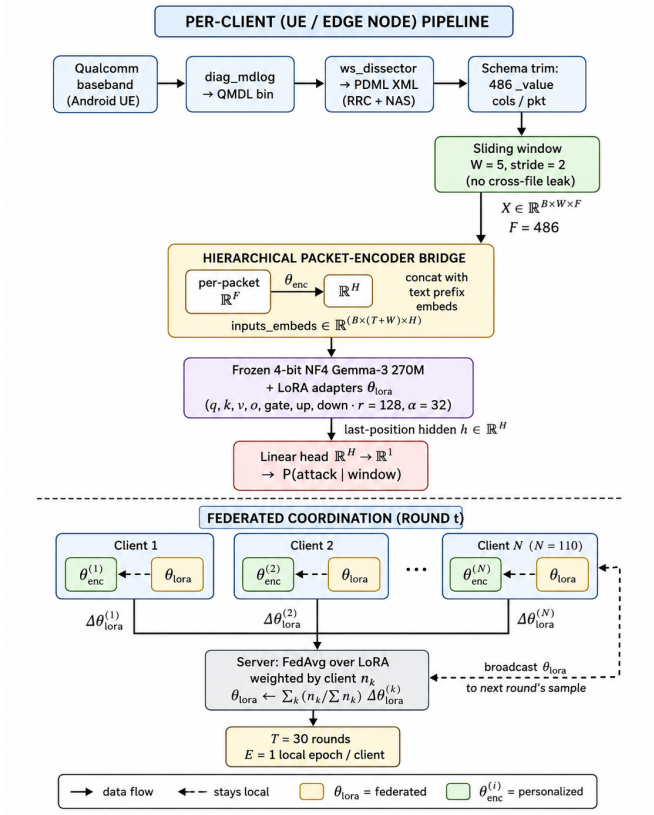


Figure 1: End-to-end methodology: capture, hierarchical encoding, frozen-SLM classification, and hierarchical federated adaptation.

natural non-IID structure of cellular captures (singleton-scenario clients, normal-only clients, attack-mix clients) without artificial Dirichlet partitioning.

3 Methodology

This study presents a decentralized framework for detecting FBS on User Equipment using a Small Language Model integrated into the 6G and B5G architecture. The overall workflow, illustrated in Fig. 1, comprises four stages: (i) capture and preprocessing of RRC and NAS signaling traces from a UE baseband, (ii) projection of each packet into the SLM's hidden space via a learned residual-MLP *packet encoder*, (iii) classification by a frozen 4-bit quantized SLM backbone with trainable LoRA adapters, and (iv) hierarchical federated training that personalizes the packet encoder per client while aggregating LoRA adapters across clients.

3.1 Dataset

3.1.1 Cellular Signaling Corpus

We evaluate the framework on the MODI cellular signaling corpus [45], which contains layer-3 PCAP captures of LTE and 5G-NR sessions collected directly from a UE baseband. The corpus comprises 49 attack PCAPs spanning 27 named scenarios across eight attack categories (encryption / security bypass, identity catching, network downgrade, service rejection, authentication abuse, location tracking, resource depletion, and stealth manipulation), together with 133 benign PCAPs captured against the same testbed with the protocol stack reconfigured per 3GPP specifications [46]. Identical radio paths between the malicious and benign corpora prevent physical-layer leakage from biasing the detector. Table 1 summarizes the attack categories covered.

3.1.2 Acquisition Pipeline

Raw diagnostic frames were captured in Qualcomm Diagnostics Logging (QMDL) format from Qualcomm-modem Android devices via the `diag_mdlog` utility while the UE established connections with both legitimate gNodeBs and rogue FBS emulations on an Open5GS + srsRAN + LibreSDR B220 testbed. QMDL binaries were converted to Packet Description Markup Language (PDML) using `ws_dissector` to isolate control-plane RRC and NAS messages, then exported to one CSV per capture with 1,949 columns per row – four standard PDML attribute variants (`_show`, `_value`, `_size`, `_pos`) per protocol field.

3.1.3 Schema Trim, Auto-Drop, and Standardization

Protocol-encoding metadata fields (`_size`, `_pos`) carry only byte offsets and lengths and are dropped. The pilot uses the numeric `_value` variant, retaining 486 columns per packet. Two further automatic filters are applied using training-set statistics only, so the same dropped-feature set is replayed on the validation and test partitions without information leakage. First, 174 columns whose standard deviation is exactly zero on the training set are removed (constant features carry no signal). Second, 23 identifier-like columns whose absolute magnitude exceeds 10^6 on the training set (TMSI, M-TMSI, MAC code, ciphered NAS blob, cell identity, and similar wide-range integer identifiers) are removed because their unbounded scale destabilizes the encoder's input projection. After auto-drop, $F = 289$ numeric features per packet remain. The retained columns are then standardized to zero mean and unit variance using per-feature training-set statistic.

3.2 Encoding Strategies: Raw-Text and Hierarchical Bridge

The standard "protocol-as-language" recipe serializes each packet's fields into a `key: value; ...` string and feeds the concatenation of W consecutive packets to a LoRA-adapted SLM. With several hundred numeric fields per packet (here, $F = 289$ after auto-drop), a $W = 10$ window reaches a 95th-percentile tokenized length of approximately 2,260 tokens and a maximum near 5,300 tokens; any reasonable `max_length` truncates the long tail of the distribution, silently discarding security-critical fields at the right end of the prompt. Shrinking the window to $W = 5$ keeps the text under 1,200 tokens but discards the temporal context required by multi-message sequence attacks.

To eliminate this token-budget ceiling we replace per-packet text serialization with a learned residual-MLP *packet encoder* that projects each packet's numeric vector into the SLM's hidden

Table 1: MODI attack-category taxonomy. The dataset's 49 attack PCAPs map onto eight categories drawn from the FBS attack-disclosure literature.

Category	Network/signaling mechanism
Encryption / security bypass	Manipulates security-mode commands or omits NAS security headers to negotiate null or weak encryption, exposing the channel to eavesdropping.
Identity catching	Collects permanent or temporary identifiers (IMSI / IMEI / SUCI / GUTI) or capability descriptors before security-context establishment.
Network downgrade	Forces the UE onto legacy or weaker-security cells via tracking-area or SIB1 manipulation, or via detach followed by redirection.
Service rejection	Issues unprotected NAS or RRC reject messages before security setup to deny attachment, tracking-area updates, or RRC re-establishment.
Authentication abuse	Triggers authentication failure or replays / malforms authentication exchanges to disrupt session establishment.
Location / tracking	Exploits measurement reports, paging procedures, or handover signaling to extract fine-grained location information.
Resource / energy depletion	Triggers repeated cryptographic procedures or authentication replays to drain UE battery.
Stealth / state manipulation	Hijacks RRC/NAS state via stealthy detach, RRC reconfiguration replay, or unsolicited EMM information messages.
Normal class – legitimate sessions, intact authentication / integrity / ciphering (133 MODI captures)	

space as a single virtual token, while retaining a short text instruction prefix so the SLM still receives natural-language framing. The encoder is summarized in Algorithm 1 and illustrated in Fig. 2: an input linear projection $R^F \to R^d$, a stack of LayerNorm–Linear–GELU–Linear residual blocks, and an output projection $R^d \to R^H$ initialized with small standard deviation (0.02) so the SLM begins close to its pretrained behavior. The encoder operates per-packet (no cross-packet attention inside the encoder); inter-packet mixing is delegated to the SLM transformer, which sees the W packet virtual tokens together with the T text instruction tokens as a single input sequence. With $W = 5$ and a $T = 24$-token prompt (Section 4.2), the SLM ingests exactly 29 tokens per forward pass.

This decomposition has three consequences. First, the SLM context length becomes constant at $T + W$ tokens regardless of window content, so truncation is eliminated by construction. Second, the SLM forward pass operates on a short sequence, recovering VRAM headroom that would otherwise be consumed by long tokenized prompts. Third, the architectural variable is cleanly ablatable: holding everything else constant and toggling between the raw-text and hierarchical encodings attributes any accuracy delta directly to the bridge.

3.3 Small Language Model Backbone

We use Gemma-3 270M [47, 48] (`gemma-3-270m-it`) as the SLM backbone. This is a compact decoder-only transformer designed for consumer-grade hardware and minimal inference latency [49], well suited to the AI-RAN paradigm where detection must occur at the network edge under tight energy budgets. The base weights are loaded in 4-bit NormalFloat (NF4) double-quantized form with bfloat16 compute [50] and held frozen throughout training; only the LoRA adapters [51] and the packet encoder are updated. The language-modeling head is replaced with a single linear classification head over the last-position hidden state, producing a binary attack/normal logit per window.

Algorithm 1 Hierarchical Packet-Encoder Encoding

Notation: $X \in R^{B\times W\times F}$ = batch of packet windows (B batch, W window length, F feature dim); $E_{\text{enc}} : R^F \to R^H$ = packet encoder; E_{tok} = SLM token embedding; p = text instruction prompt; H = SLM hidden dim.

Input: packet windows X, instruction prompt p
Output: `inputs_embeds` $\in R^{B\times(T+W)\times H}$ for the frozen SLM

$e_{\text{txt}} \leftarrow E_{\text{tok}}(\text{tokenize}(p))$ $\{R^{T\times H}$, broadcast over batch$\}$
for $b = 1$ to B **do**
  for $w = 1$ to W **do**
    $e_{\text{pkt}}[b, w] \leftarrow E_{\text{enc}}(X[b, w])$
  end for
end for
`inputs_embeds` $\leftarrow \text{concat}(e_{\text{txt}}, e_{\text{pkt}}, \text{axis} = 1)$
return `inputs_embeds`

3.4 Hierarchical Federated Adaptation

Naively federating the entire detector treats the packet encoder and the LoRA adapters as a single payload, even though they capture fundamentally different signal. The packet encoder absorbs device-level feature statistics that vary across UEs (baseband firmware, modem-vendor field encodings, capture-time scaling), while the LoRA adapters carry protocol-violation reasoning that should be shared across all clients in order to generalize. We exploit this distinction in *Hierarchical Federated Adaptation* (HFA): the packet encoder $\theta_{\text{enc}}^{(i)}$ is maintained *per client* i and never aggregated, while the LoRA adapters θ_{lora} are aggregated across clients via FedAvg weighted by client local sample counts. The frozen 4-bit base SLM is broadcast once during initialization and never crosses the wire afterwards.

To isolate the effect of this decomposition we define four training regimes, each evaluated with otherwise-identical data splits, hyperparameters, and seeds: (i) **centralized**, a single global model trained on the pooled data as an upper-bound reference; (ii) **FedAvg-full**, FedAvg over both the packet encoder and the LoRA adapters; (iii) **FedAvg-LoRA**, FedAvg over LoRA only with the encoder weights kept at their (optionally centrally warmed-up) initialization across clients; and (iv) **HFA**, FedAvg over LoRA with each client maintaining its own personalized packet encoder. The architectural argument is that regime (iii) tests whether encoder federation is load-bearing, and regime (iv) tests whether personalization recovers any accuracy lost in (iii) while still benefiting from cross-client LoRA aggregation.

This orchestration ensures that sensitive user identity, location, and capability data never leave the device, providing a privacy-by-design security layer that is robust against localized FBS variants.

4 Experiment

4.1 Data Splits and Client Partition

Splits operate at the *file* level rather than the row level: partitioning rows after sliding-window construction would let train and test windows share underlying packets, a leakage path that prior FBS-detection work has not consistently addressed. The 133 benign files are partitioned

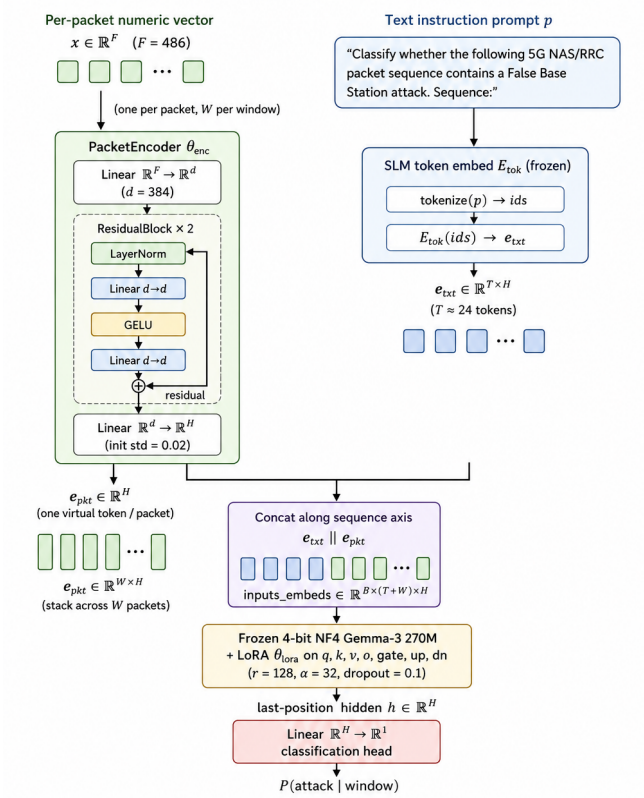


Figure 2: Hierarchical packet-encoder bridge: each packet's numeric feature vector is projected by a residual-MLP encoder into a single virtual token that is concatenated with the text instruction prefix embeddings before entering the frozen 4-bit SLM.

via a seeded 70/20/10 random split. Multi-file attack scenarios use a per-scenario seeded shuffle with rounded 70/20/10 quotas so every scenario appears in training whenever it has more than one capture.

Sliding windows are constructed inside each file with $W = 5$ and stride 2; windows never cross file boundaries. Within each split, the majority class is downsampled to enforce a 1:1 class balance. Table 2 summarizes the resulting partition for seed 13. For federation, the training-set files become the federated clients: each file is one client (*per-file* client granularity), giving a naturally non-IID benchmark of 110 clients in which many singleton-scenario clients see only one attack family. A per-client window cap of 300 prevents large captures (e.g., the multi-thousand-window `panic_attack` file) from dominating the FedAvg aggregation; in the centralized baseline, the same training pool is capped at 3,500 windows so the centralized and federated regimes consume comparable budgets.

4.2 Implementation Details

Backbone. The detection engine is `gemma-3-270m-it` [47, 48], loaded in 4-bit NF4 double-quantized form with bfloat16 compute via the bitsandbytes integration of QLoRA [50]. The base weights are frozen; only the LoRA adapters, packet encoder, and classification head are

Table 2: File-Level Data Partition After Scenario Exclusion (seed 13).

Partition	**Files**	**Clients (FL)**	**Windows (1:1 balanced)**
Training	110	110 (per-file)	$\leq$300/client; $\leq$3,500 pooled
Validation	34	–	$\sim$1,500
Test	20	–	$\sim$1,280

trainable, totaling approximately 11M parameters or roughly 4% of the base model.

Adapters and encoder. LoRA [51] adapters are applied to the attention and MLP projections (`q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, `down_proj`) with rank $r = 128$, scaling $\alpha = 32$, and dropout 0.1. The packet encoder uses inner dimension $d = 384$ and two residual blocks (dropout 0.1), contributing approximately 1.6M trainable parameters; the linear classification head over the last-position hidden state contributes 640 additional parameters.

Instruction prompt. The fixed text prefix concatenated with the W packet virtual tokens is the 24-token string ``` ``Classify whether the following 5G NAS/RRC packet sequence contains a False Base Station attack. Sequence:'' ```, yielding a constant SLM context length of $T + W = 29$ tokens per forward pass.

Federated round structure. The experiment configuration runs $R = 20$ communication rounds with $K = 6$ clients sampled uniformly at random per round (approximately 5.5% of the 110 train clients) and $E = 1$ local epoch per sampled client. Local optimization uses AdamW with learning rate $\eta = 2 \times 10^{-4}$ and weight decay 0.01, per-device batch size 1, and gradient accumulation of 4 (effective batch 4). Aggregation is FedAvg weighted by per-client local sample counts. For the centralized baseline, the product $R \cdot E = 20$ is interpreted as the number of epochs over the pooled 3,500-window training set, so all four regimes traverse a comparable number of parameter updates.

Modes. Four training regimes are run under otherwise-identical data splits, seed, and hyperparameters, differing only in which parameter groups participate in federation: *centralized* (no FL loop), *FedAvg-full* (encoder + LoRA aggregated), *FedAvg-LoRA* (LoRA aggregated, encoder frozen at random initialization across all clients), and *HFA* (LoRA aggregated, encoder personalized and cached per client between rounds). The frozen 4-bit base never crosses the wire; the per-round payload comprises only the aggregated parameter groups. For HFA global evaluation, the global LoRA is paired with the mean of all currently cached per-client encoders.

Evaluation protocol. Held-out test and validation metrics are evaluated every four rounds (five evaluation points per run plus a final point); HFA additionally reports per-client local test accuracy at the final round, with each client's personalized encoder paired with the global LoRA on a 20% held-out portion of that client's own data. We report raw TP/FP/TN/FN counts, accuracy, precision, recall, F1, and FPR.

Hardware and software. Experiments are conducted on a single NVIDIA GeForce RTX 4080 GPU (16 GB VRAM) hosted by an Intel Core i7-13700K CPU (3.40 GHz) with 64 GB of system RAM, running 64-bit Ubuntu 24.04 LTS. The implementation uses Python 3.10, PyTorch 2.9.1 with CUDA 12.8.

4.3 Evaluation Metrics

To evaluate the performance of classification models, we adopt a range of metrics derived from the confusion matrix. This matrix provides a summary of prediction outcomes by comparing actual class labels with those predicted by the model, as illustrated in Table 3. In the context of binary classification, we categorize the outcomes into four primary types: True Positives (TP), where the model correctly classifies positive instances; False Positives (FP), where the model incorrectly predicts negative instances as positive (Type I error); True Negatives (TN), where the model correctly identifies negative instances; and False Negatives (FN), where the model mistakenly classifies positive instances as negative (Type II error).

Table 3: Confusion Matrix

	Predicted as Positive	Predicted as Negative
Labeled as Positive	TP	FN
Labeled as Negative	FP	TN

Accuracy (ACC) is the proportion of correct predictions among all predictions, providing an overall sense of model performance. Precision (PR) focuses on the proportion of predicted positive cases that are actually positive, indicating the accuracy of positive predictions. Recall (RC) measures the model's ability to correctly identify actual positive cases, indicating how well it captures all positives. False Positive Rate (FPR), on the other hand, indicates the proportion of actual negatives that were incorrectly classified as positive, reflecting the model's tendency to generate false alarms.

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$RC = \frac{TP}{TP + FN} \tag{2}$$

$$FPR = \frac{FP}{FP + TN} \tag{3}$$

$$PR = \frac{TP}{TP + FP} \tag{4}$$

$$F1\ Score = \frac{2 * PR * RC}{PR + RC} \tag{5}$$

5 Results and Discussion

We report preliminary experiment results across all four training regimes defined in Section 3.4 – *centralized*, *FedAvg-full*, *FedAvg-LoRA*, and *HFA* – at seed 13, on the held-out test partition described in Section 4.1 ($n = 1{,}282$ balanced windows).

5.1 Detection Performance

Table 4 reports the final-round detection performance for each regime, with Wilson 95% confidence intervals on accuracy. Four observations follow.

First, all three federated regimes match or exceed the centralized upper bound (FedAvg-full F1 +0.7 pp; FedAvg-LoRA +1.6 pp; HFA +1.4 pp), supporting the privacy-by-design framing: federation incurs no detection-quality penalty on this pilot relative to pooling raw signaling logs centrally.

Second, FedAvg-LoRA matches and slightly exceeds FedAvg-full (ACC +0.9 pp, F1 +0.9 pp), with both models trained from identical data, splits, and seed. This directly addresses the encoder-not-load-bearing question: excluding the packet encoder from the FedAvg payload imposes no measurable accuracy cost in this regime, and the aggregated encoder appears to add a slight source of cross-client drift that the LoRA-only variant avoids.

Third, HFA shifts the operating point toward fewer false alarms. While HFA's accuracy and F1 are statistically indistinguishable from FedAvg-LoRA (ACC 0.832 vs. 0.833; F1 0.848 vs. 0.850), HFA achieves the highest precision (0.774) and the lowest false-positive rate (0.275) of any regime evaluated, including the centralized baseline. This is the empirical signature that personalizing the packet encoder absorbs client-level feature variation: each client's encoder adapts to its own `_value`-column distribution, so the LoRA layers above are presented with more consistent inputs and fire fewer spurious attack predictions on benign traffic.

Fourth, all regimes exhibit a high-recall / moderate-precision profile (RC $\in [0.914, 0.949]$, PR ≈ 0.77, FPR $\in [0.275, 0.284]$), with FedAvg-LoRA achieving the lowest false-negative count (FN = 33). For the FBS detection use case this error asymmetry is operationally desirable: a missed FBS leaves the UE exposed to identity catching or downgrade for the duration of the false attachment, whereas a false alarm can be reconciled downstream by a complementary specification-based check [10] or by per-RAN-cell whitelisting before any user-visible action is taken.

Table 4: Final-round detection performance on the held-out test partition

Mode	ACC	PR	RC	F1	FPR
Centralized	0.818	0.767	0.914	0.834	0.278
FedAvg-full	0.824	0.767	0.933	0.842	0.284
FedAvg-LoRA	**0.833**	0.771	**0.949**	**0.850**	0.282
HFA (proposed)	0.832	**0.774**	0.938	0.848	**0.275**

5.2 Communication Cost

Per-round communication cost is deterministic given the architecture: a bf16-serialized LoRA adapter state under the configuration of Section 4.2 is approximately 58 MB, and the packet encoder state is approximately 1.6 MB. Table 5 reports the resulting per-round and pilot-total payload across the four regimes, computed as $K \cdot 2 \cdot$ payload per round (factor of two for the download + upload halves of FedAvg) and multiplied by $R = 20$ rounds. As a reference point, federating the full bf16 base model (no quantization, no PEFT) would transmit approximately 540 MB per client per direction, yielding roughly 130 GB over the same pilot, so LoRA-only federation reduces communication by approximately one order of magnitude. The packet encoder

contributes under 3% of the FedAvg-full payload, which already foreshadows the FedAvg-LoRA vs. FedAvg-full accuracy. If the encoder is not load-bearing for federation, the 3% saving comes at no cost.

Table 5: Deterministic per-round and experiment-total communication cost (bf16, $K = 6$ clients per round, $R = 20$ rounds).

Mode	**Payload / client / dir**	**Per-round (MB)**	**Pilot total (MB)**
FedAvg-full	LoRA + encoder ≈ 59.6 MB	≈ 715	≈ 14,300
FedAvg-LoRA	LoRA ≈ 58 MB	≈ 696	≈ 13,920
HFA (proposed)	LoRA ≈ 58 MB	≈ 696	≈ 13,920
Full bf16 base (ref.)	*≈ 540 MB*	*≈ 6,480*	*≈ 129,600*

5.3 Discussion

The preliminary experiment supports all three claims targeted by this work, with different strength along each axis. First, federation does not cost accuracy on this benchmark: all three federated regimes match or exceed the centralized upper bound (FedAvg-LoRA F1 0.850 vs. centralized 0.834), confirming that the on-device federated training regime is viable under the privacy constraints that motivated it. Second, the packet encoder is not load-bearing for federation: FedAvg-LoRA matches FedAvg-full at lower per-round payload (Table 5), so the simpler, cheaper aggregation strategy is the right default in deployments where the encoder can be initialized once and held fixed across clients. Combined with the architectural reduction from federating a full bf16 base, the total per-round communication of FedAvg-LoRA is roughly one order of magnitude lower than naïve full-model federation.

Third, personalizing the packet encoder under HFA does not improve aggregate accuracy on this single-testbed pilot, but it does shift the operating point toward fewer false alarms: HFA matches FedAvg-LoRA on accuracy and F1 within 0.2 pp while achieving the highest precision (0.774) and the lowest false-positive rate (0.275) of any regime tested, including the centralized baseline. This is the empirical signature of the personalization hypothesis: each client's encoder absorbs that client's idiosyncratic feature distribution, so the LoRA layers receive a more consistent input across the federation and fire fewer spurious attack predictions on benign traffic. We therefore frame HFA as the right choice when false-alarm cost dominates (e.g., when each alert pages an operator or triggers a downstream verification) and FedAvg-LoRA as the right choice when recall dominates (e.g., when the cost of a missed FBS exceeds the cost of a false alarm by a wide margin).

Finally, the high-recall / moderate-precision profile observed across all four regimes is operationally favorable for an FBS detector but does warrant a downstream filtering layer in deployment. Pairing the SLM with a specification-based check [10] or with a CellGuard-style location-database cross-check [36] would let the SLM run with high recall as a first-stage trigger while the downstream layer absorbs the elevated false-positive rate before any user-visible action is taken.

6 Conclusion

The necessity for a decentralized intelligence architecture in 6G stems from the logistical and regulatory difficulties inherent in modern raw data management. This paper introduces an on-device FBS detection framework that pairs a hierarchical packet-encoder bridge with a frozen 4-bit quantized Small Language Model, projecting each RRC/NAS packet into the SLM's hidden space as a single virtual token and eliminating the token-budget ceiling imposed by raw-text serialization. We further introduce Hierarchical Federated Adaptation (HFA), a federation strategy that personalizes the packet encoder per client to absorb device-level feature variation while aggregating LoRA adapters across clients via FedAvg, allowing each client to fit its local feature distribution without sacrificing the cross-client protocol reasoning carried by the shared adapters. By utilizing the Gemma-3 270M backbone together with 4-bit NF4 quantization and LoRA-based parameter-efficient fine-tuning, the framework targets the stringent VRAM and latency envelopes of edge nodes. Preliminary pilot results on the MODI cellular signaling corpus across all four training regimes substantiate the work's three claims: federation incurs no detection-quality penalty (FedAvg-LoRA test F1 0.850 vs. centralized 0.834); excluding the packet encoder from the federated payload imposes no measurable accuracy cost while reducing per-round communication by approximately one order of magnitude relative to federating the full bf16 base; and personalizing the packet encoder under HFA shifts the operating point toward fewer false alarms (FPR 0.275, the lowest of any regime tested) at parity F1 with FedAvg-LoRA. Multi-seed Wilson confidence intervals, per-client local-test breakdowns, and per-attack-scenario recall on the OOD-heavy validation partition will accompany the extended version of this work; this work establishes the architectural scaffolding for a scalable, privacy-by-design security layer aligned with the AI-RAN paradigm in the B5G and 6G era.

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